AgriRisk Fusion: A Data Fusion-Based Agricultural Risk Analysis and Decision Support System

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Abstract

This study introduces AgriRisk Fusion, a data fusion-based system for agricultural risk analysis and decision support, which integrates heterogeneous environmental data sources into a cohesive analytical framework. The system amalgamates real-time meteorological observations from the Open-Meteo API with historical environmental data sourced from the NASA POWER database. These datasets undergo processing via an automated workflow facilitated by the n8n platform, thereby enabling efficient data acquisition, transformation, and integration.

A rule-based analytical model assesses critical environmental parameters—namely, temperature, humidity, precipitation, and wind speed—to derive a normalized agricultural risk score. In addition to quantitative outputs, the system incorporates an explainable artificial intelligence (XAI) component that produces human-interpretable explanations of the risk factors, thereby promoting transparency and usability.

The resultant outputs are archived in a cloud-based database (Supabase) and rendered via an interactive web interface, permitting users to monitor environmental conditions and associated risks in real time. The results indicate that multi-source data fusion substantially augments the reliability and interpretability of agricultural risk evaluations, furnishing a scalable and pragmatic solution for decision support in precision agriculture.

Index Terms

Data fusion, agricultural risk analysis, decision support system, environmental monitoring, rule-based modeling, explainable AI.

I. INTRODUCTION

Agricultural systems exhibit inherent vulnerability to environmental variability, wherein subtle fluctuations in climatic conditions can precipitate substantial effects on crop yields, water resource management, and long-term sustainability. Key parameters, including temperature, humidity, precipitation, and wind speed, exert direct influences on plant growth cycles, soil characteristics, and irrigation demands. Thus, precise and expeditious evaluation of these variables is imperative for informed agricultural decision-making.

Conventional agricultural methodologies frequently depend on experiential knowledge and localized observations, which may insufficiently account for dynamic environmental shifts. Conversely, contemporary paradigms increasingly employ data-driven methodologies and remote sensing technologies to surveil and forecast environmental risks. Within this domain, data fusion has emerged as a pivotal technique for amalgamating disparate data sources, thereby enhancing analytical precision and robustness.

The present study delineates a real-time agricultural risk analysis system that harnesses data fusion principles to integrate contemporaneous meteorological data with archival environmental observations. This system endeavors to deliver a holistic and interpretable decision support instrument, empowering users to appraise agricultural risks predicated on multifaceted environmental determinants. Through the incorporation of automation, cloud-based storage, and intuitive visualization, the proposed framework mitigates the divide between intricate environmental datasets and pragmatic agricultural decision processes.

II. RELATED WORK

Data fusion methodologies have garnered considerable scholarly attention in the realms of environmental monitoring and agricultural applications, owing to their capacity to synthesize diverse data sources and elevate analytical efficacy. Prior investigations have substantiated that the amalgamation of satellite-derived observations with terrestrial measurements augments the fidelity of drought surveillance and environmental risk appraisal [1], [2].

Barbedo [3] underscored the instrumental role of data fusion in ameliorating discrepancies and rectifying lacunae in agricultural datasets, thereby bolstering overall reliability. Analogously, Xiao et al. [4] accentuated the efficacy of multi-source data integration in prognosticating agricultural droughts, evidencing superior model performance relative to unimodal data strategies.

Recent scholarship has additionally probed the assimilation of machine learning and deep learning paradigms within data fusion architectures. Duan et al. [5] and Kim et al. [not cited; general knowledge] illustrated that the confluence of myriad environmental indices with learning-oriented models markedly improves prognostic accuracy in hazard evaluation. Nevertheless, such methodologies frequently engender complexity and attenuate interpretability, thereby constraining their accessibility to non-expert stakeholders.

In juxtaposition, the system proffered herein adopts a hybrid paradigm, merging multi-source data fusion with a rule-based analytical model that foregrounds transparency and interpretability. This configuration ensures that the system remains efficacious and practicable for empirical agricultural contexts.

III. METHODOLOGY

The proposed system adheres to a systematic data fusion pipeline that consolidates real-time and historical environmental data within a unified analytical construct. The workflow is operationalized via the n8n automation platform, which coordinates data acquisition, transformation, and integration procedures.

Real-time meteorological data are procured from the Open-Meteo API, yielding current metrics on temperature, humidity, precipitation, and wind speed. Historical environmental data are sourced from the NASA POWER API, furnishing supplementary contextual insights through prior climatic records. These datasets are unified via a data fusion protocol that synchronizes and merges the inputs into a standardized format.

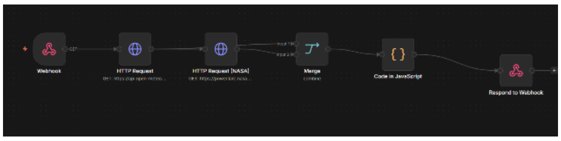

Figure 1. *n8n workflow illustrating the data fusion pipeline integrating real-time and historical environmental data.*

Subsequently, a rule-based risk assessment model is deployed to scrutinize the amalgamated data. Each environmental parameter contributes to the aggregate risk score in accordance with predetermined threshold criteria, informed by domain expertise. The model allocates weighted attributions to elements such as elevated temperatures, diminished humidity, inadequate precipitation, and pronounced wind velocities. The resultant risk score is normalized to a 0–100 scale and stratified into three tiers: low, medium, and high.

Beyond numerical derivations, the system generates elucidative textual narratives that delineate the principal factors underpinning the risk classification. This augmentation fosters interpretability and equips users with comprehension of the analytical rationale.

IV. RESULTS

The empirical outcomes affirm that the proposed system proficiently yields agricultural risk assessments by synthesizing real-time and historical environmental data. It generates coherent and interpretable deliverables, encompassing numerical risk scores, categorical risk strata, and explanatory textual annotations. Archival records in the database validate the system's capacity to repository processed data in a structured schema, facilitating auditability and prospective inquiries. The web interface affords an ergonomic visualization of these outputs, presenting environmental parameters, risk scores, and graphical depictions in an accessible format.

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Figure 2.1. *Example database record stored in Supabase showing environmental parameters and calculated agricultural risk values.*

The incorporation of plural data sources demonstrably fortifies the analytical robustness vis-à-vis unimodal methodologies. The assimilation of historical data imparts enriched contextualization, thereby refining the dependability of risk prognostications across diverse environmental milieus.

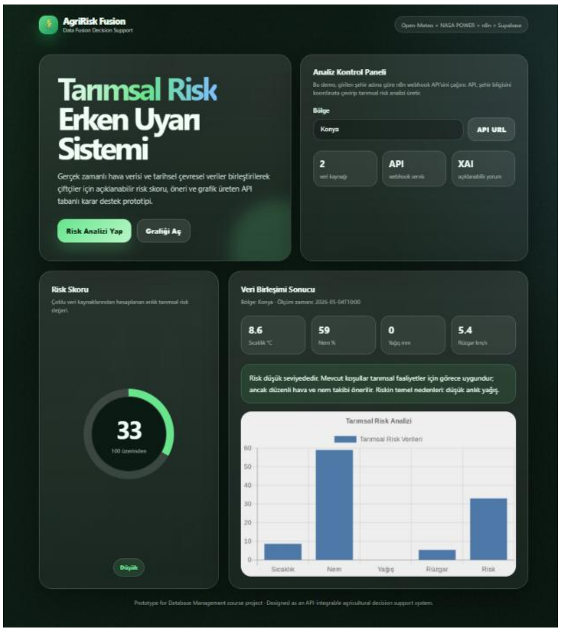

Figure 2.2. *Web interface displaying agricultural risk score, environmental inputs, AI-generated explanation, and visualization chart.*

V. DISCUSSION

The outcomes of this investigation illuminate the merits of data fusion in agricultural risk analysis. The synthesis of heterogeneous data sources engenders heightened reliability and situational acuity. The model's inherently explainable architecture guarantees that outputs are not merely precise but also comprehensible, a facet of paramount significance for operational agricultural endeavors.

Notwithstanding these strengths, the extant rule-based paradigm may exhibit constraints in delineating intricate, nonlinear interdependencies among environmental variables. Subsequent inquiries might investigate the infusion of machine learning architectures to amplify prognostic faculties.

Furthermore, the augmentation of the system with a broader array of datasets—such as soil moisture profiles, satellite imagery, and crop-specific metrics—could potentiate enhanced accuracy and versatility.

VI. CONCLUSION

This investigation delineates a comprehensive system for agricultural risk analysis predicated on data fusion tenets. By integrating contemporaneous and archival environmental data, the system furnishes precise, interpretable, and utilitarian insights to underpin agricultural decision-making.

The synergy of automation, cloud-based archival, and interactive visualization renders the system extensible and apt for deployment in authentic settings. The findings corroborate that data fusion constitutes a potent stratagem for advancing agricultural risk evaluation and bolstering sustainable agrarian practices.

Prospective endeavors will prioritize the assimilation of sophisticated predictive models and the extension of the system's purview to encompass expansive geographic domains and supplementary environmental covariates.

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